Log

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The program's notebook — the *why* and *what-happened* that the run-stamps do not capture. Append-only, newest first. Relaxed and chronological by design; still a published page, so run numbers cite the record, never hand-typed. The goal and the queue live in the plan the operating policy in the night-shift.

Digest

Latest, for the 30-second morning read.

Program opened. Goal set: find the minimal recipe that trains a conductance E/I net stably on SHD. exp060 smoke trains to 61% on a subset but **diverges intermittently** (NaN forward-passes, unbounded W_ee). Stability queue (dt / Dale's law / weight decay) is registered and waiting. **Open anomaly for a human:** NaN appears even at epoch 2 when weights are small — so it is not weight-growth runaway; likely integration stiffness at dt = 1.0.

Sessions

2026-07-11 — program opened

Stood the program up end to end in one sitting.

- **Built the substrate.** SHD training now works through the CLI: an event-based loader (official train/test split, lazy binning to the model's dt), all four recurrent blocks made trainable (the W_ii CLI path was a latent dead flag, now fixed), and an AdamW weight-decay knob. The data-look entry confirmed the dataset is clean and class-separable.
- **Ran the first training smoke** (exp060, Rung A: free signed-recurrent ceiling, dt 1.0, 1000-sample subset). It trains — loss 3.18 → 0.56, best test accuracy 61% against a 5% floor — so the pipeline is sound. But the free dynamics are unstable: scattered epochs return NaN train/test loss, pre-clip gradient norms spike to millions, and the E→E recurrent weight grows without bound.
- **Chased the instability and was wrong once.** First guess was unbounded weight growth; added weight decay. It tamed the gradient explosion (7.5M → 900) and lifted final accuracy (50% → 58%), but did *not* bound W_ee at 1e-3 and made the NaN *more* frequent — and a NaN showed up at epoch 2 with tiny weights. So the root cause is not weight size; it is an intermittent forward-pass divergence of the signed conductance dynamics, most likely exp-Euler stiffness at coarse dt. Parked for exp061.
- **Reframed the program.** What began as debugging is the actual goal: a stable training recipe, with each ingredient attributed. Queued exp061 (dt sweep), exp062 (Dale's law as the implicit stabiliser), exp063 (weight-decay sweep), with a forward-state clamp held in reserve.